

Littlewood’s Problem 22 on the Zeros of Cosine Polynomials: A Candidate Improved Upper Bound $Z(N) \ll N^{2/3}(\log N)^{1/3}$

Hu yuanshan *

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Abstract

Let $Z(N)$ denote the minimum, over all cosine polynomials $F_A(x) = \sum_{a \in A} \cos(ax)$ with $A \subset \mathbb{Z}_{\geq 0}$, $|A| = N$, of the number of distinct real zeros of F_A in $[0, 2\pi)$. Konyagin [1] and, independently, Juškevičius and Sahasrabudhe [2] proved the upper bound $Z(N) \ll N^{2/3}(\log N)^{2/3}$. We present a self-contained proof of a candidate improvement

$$Z(N) \ll N^{2/3}(\log N)^{1/3},$$

which saves a factor $(\log N)^{1/3}$ over the previously known bound. The argument has two independent parts: a probabilistic construction of a low-frequency 0/1 cosine polynomial whose “small-band” set has small measure, and a deterministic argument counting the zeros of a perturbation of the Dirichlet kernel. The result should be regarded as a *candidate* theorem: the proof has passed an internal multi-agent audit but has not been confirmed by external experts or by a formal proof system.

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1 Introduction

For a finite set $A \subset \mathbb{Z}_{\geq 0}$ write

$$F_A(x) = \sum_{a \in A} \cos(ax),$$

and define

$$Z(N) = \min_{|A|=N} \#\{x \in [0, 2\pi) : F_A(x) = 0\},$$

where zeros are counted as *distinct* points (without multiplicity). Littlewood’s Problem 22 asks for the order of growth of $Z(N)$. The strongest upper bound currently in the literature is

$$Z(N) \ll N^{2/3}(\log N)^{2/3},$$

proved independently by Konyagin [1] and by Juškevičius and Sahasrabudhe [2]. The purpose of this note is to give a self-contained proof of the following candidate improvement.

*Replace with the author’s name and affiliation.

Theorem 1. *For all sufficiently large N ,*

$$Z(N) \ll N^{2/3}(\log N)^{1/3}.$$

Relative to the bound $Z(N) \ll N^{2/3}(\log N)^{2/3}$, Theorem 1 saves a factor $(\log N)^{1/3}$.

Remark 2 (Status of the proof). The proof below is a single linear, self-contained argument. Each auxiliary estimate has been checked by an internal multi-agent audit (adversarial re-derivation of the lemmas, together with a numerical search for small-parameter counterexamples); however, the result has not yet been verified by an external expert, by a literature-novelty search, or by a formal proof system. It should therefore be treated as a *candidate* improvement rather than an established theorem. AI tools were used in the drafting and verification of this manuscript (see the Acknowledgements).

The proof has two independent parts. The first is a probabilistic construction of a low-frequency 0/1 cosine polynomial g whose small-band set

$$E(g) = \left\{ x \in (0, \pi) : \left| 2 \sin(x/2) \left(g(x) - \frac{1}{2} \right) \right| \leq 1 \right\}$$

has small measure, the precise target being

$$|E(g)| \ll \sqrt{\frac{\log m}{m}},$$

where m is a parameter controlling the degree of g . The second part is a deterministic counting argument: if g has degree at most $m < n$, then

$$Z(D_n - g) \ll n |E(g)| + m + 1,$$

where $D_n(x) = \sum_{k=0}^n \cos(kx)$ is the Dirichlet kernel. Balancing the two contributions with $m \asymp N^{2/3}(\log N)^{1/3}$ yields Theorem 1.

2 Notation and conventions

Throughout, $s = \sin(x/2)$. The symbol $\ll$ denotes the Vinogradov bound $|f| \leq C|g|$ for an absolute constant C ; all unspecified constants are absolute. We write $\log$ for the natural logarithm. For a measurable set $S \subset \mathbb{R}$, $|S|$ denotes its Lebesgue measure, while for a finite set A , $|A|$ denotes its cardinality; the meaning will always be clear from the context. For a cosine polynomial f , $Z(f)$ denotes the number of distinct zeros of f in $[0, 2\pi)$; thus $Z(N) = \min_{|A|=N} Z(F_A)$. We only prove the result for sufficiently large N ; finitely many small values are absorbed into the implied constant.

3 The Dirichlet-kernel reduction

Lemma 3 (Dirichlet kernel identity). *For $x \notin 2\pi\mathbb{Z}$,*

$$D_n(x) = \frac{1}{2} + \frac{\sin((n + \frac{1}{2})x)}{2 \sin(x/2)}.$$

Proof. Using

$$2 \sin(x/2) \cos(kx) = \sin((k + \frac{1}{2})x) - \sin((k - \frac{1}{2})x),$$

sum over $0 \leq k \leq n$. The intermediate terms cancel in pairs, leaving

$$2 \sin(x/2) D_n(x) = \sin((n + \frac{1}{2})x) + \sin(x/2).$$

Dividing by $2 \sin(x/2)$ gives the claim. $\square$

Now let g be a 0/1 cosine polynomial of degree at most $m < n$, and set

$$h(x) = 2 \sin(x/2)(g(x) - \frac{1}{2}), \quad M = n + \frac{1}{2}.$$

By Lemma 3,

$$D_n(x) - g(x) = 0 \iff \sin(Mx) = h(x).$$

Consequently zeros of $D_n - g$ can occur only where $|h(x)| \leq 1$; this is exactly the set $E(g)$ defined in the Introduction.

4 A random block model

Fix a large integer m . Choose integers

$$8 \leq L \leq m^{1/3}, \quad q = \left\lfloor \frac{m+1}{4L} \right\rfloor.$$

Partition the frequency axis into q boxes of length $4L$. For each $j \in \{0, \dots, q-1\}$, independently choose

$$U_j \sim \text{Unif}\{0, \dots, L-1\}, \quad R_j \sim \text{Unif}\{L, \dots, 2L-1\},$$

and define the random interval

$$I_j = \{4Lj + U_j, \dots, 4Lj + U_j + R_j - 1\}.$$

Since $U_j + R_j - 1 \leq 3L - 2 < 4L$, the interval I_j lies entirely inside the j -th box; the intervals I_j are therefore pairwise disjoint. Independently choose

$$\eta_j \sim \text{Bernoulli}(1/2)$$

and set

$$g(x) = \sum_{j=0}^{q-1} \eta_j \sum_{k \in I_j} \cos(kx).$$

Because the intervals are disjoint, every coefficient of g lies in $\{0, 1\}$. Moreover the right endpoint of the last interval is at most

$$4L(q-1) + (L-1) + (2L-1) - 1 = 4Lq - L - 3 \leq m,$$

so $\deg g \leq m$.

For fixed j, u, r , write

$$b_{j,u,r}(x) = \sum_{k=4Lj+u}^{4Lj+u+r-1} \cos(kx), \quad s = \sin(x/2).$$

Lemma 4 (Exact block sum). *For $0 < x < 2\pi$,*

$$2s b_{j,u,r}(x) = 2 \sin(rx/2) \cos\left(4Ljx + \frac{(2u+r-1)x}{2}\right).$$

Proof. By the finite geometric sum

$$\sum_{k=a}^{a+r-1} e^{ikx} = e^{i(a+(r-1)/2)x} \frac{\sin(rx/2)}{\sin(x/2)},$$

take real parts with $a = 4Lj + u$. □

Lemma 4 shows that the natural scale of a block is

$$A_L(x) = \min\{L, 1/s\}.$$

5 A length-energy lemma

Lemma 5 (Square mean over contiguous block lengths). *There is an absolute constant $c_A > 0$ such that for all $L \geq 2$ and all $0 < t \leq \pi/2$,*

$$\frac{1}{L} \sum_{r=L}^{2L-1} \sin^2(rt) \geq c_A \min\{(L \sin t)^2, 1\}.$$

Proof. Using $\sin^2 y = (1 - \cos 2y)/2$ and the finite cosine-sum formula,

$$\frac{1}{L} \sum_{r=L}^{2L-1} \sin^2(rt) = \frac{1}{2} - \frac{\sin(Lt)}{2L \sin t} \cos((3L-1)t).$$

Suppose first that $L \sin t \geq 2$. Then the error term has absolute value at most $1/4$, so the left-hand side is at least $1/4$.

Assume henceforth that $L \sin t < 2$. Since $\sin t \geq 2t/\pi$, we have $Lt < \pi$.

If $Lt \leq \pi/4$, then for $L \leq r \leq 2L-1$ we have $0 < rt \leq \pi/2$, whence

$$\sin(rt) \geq \frac{2rt}{\pi} \geq \frac{2Lt}{\pi}.$$

Therefore

$$\frac{1}{L} \sum_{r=L}^{2L-1} \sin^2(rt) \geq \frac{4}{\pi^2} L^2 t^2 \geq \frac{4}{\pi^2} (L \sin t)^2.$$

Finally suppose $\pi/4 < Lt < \pi$. The function

$$t \mapsto \frac{\sin(Lt)}{L \sin t}$$

is decreasing on $(0, \pi/L)$, because $z \cot z$ is strictly decreasing on $(0, \pi)$. Hence

$$0 \leq \frac{\sin(Lt)}{L \sin t} \leq \frac{\sin(\pi/4)}{L \sin(\pi/(4L))} \leq \cos(\pi/8) < 0.924.$$

Substituting this into the exact formula above, the left-hand side is at least

$$\frac{1 - \cos(\pi/8)}{2} > 0.038.$$

Combining the three cases yields the lemma; for instance one may take $c_A = 10^{-3}$. □

6 The resonance layer and block abundance

Fix $D = 4$ and assume throughout this section that $q \geq 8$. Define the resonance layer

$$\mathcal{R} = \left\{ x \in [0, \pi] : |\sin(4Lx)| \leq D/q \right\}.$$

Lemma 6 (Measure of the resonance layer). *For all $q \geq 1$,*

$$|\mathcal{R}| = 2 \arcsin(\min\{1, D/q\}) \leq \frac{\pi D}{q}.$$

In particular, if $q \geq 2D$, then

$$|\mathcal{R}| = 2 \arcsin(D/q) \leq 4D/q.$$

Proof. Put $y = 4Lx$. As x runs over $[0, \pi]$, y runs over $[0, 4L\pi]$, which comprises exactly $4L$ periods of length π of $|\sin y|$. For $0 \leq \delta \leq 1$ the set $\{|\sin y| \leq \delta\}$ has length $2 \arcsin \delta$ in each period. The factor $4L$ in the change of variables cancels with the number of periods, giving the first equality. The inequality uses $\arcsin z \leq (\pi/2)z$ for $0 \leq z \leq 1$; for $z \leq 1/2$ one may also use the convenient bound $\arcsin z \leq 2z$. $\square$

Lemma 7 (Non-resonant phase average). *If $x \notin \mathcal{R}$, then for every real α ,*

$$\frac{1}{q} \sum_{j=0}^{q-1} \cos^2(4Ljx + \alpha) \geq \frac{3}{8}.$$

Proof. Since $\cos^2 y = (1 + \cos 2y)/2$,

$$\frac{1}{q} \sum_{j=0}^{q-1} \cos^2(4Ljx + \alpha) = \frac{1}{2} + \frac{1}{2q} \Re \left(e^{2i\alpha} \frac{1 - e^{i8Lqx}}{1 - e^{i8Lx}} \right).$$

The geometric-sum error term has absolute value at most

$$\frac{1}{2q |\sin(4Lx)|} < \frac{1}{2D} = \frac{1}{8}.$$

The claim follows. $\square$

Write

$$\Delta(x) = \min\{Ls, 1\}, \quad Y_{j,u,r}(x) = 2s b_{j,u,r}(x).$$

Lemma 8 (A fixed proportion of good triples). *There are absolute constants $c_1, \gamma > 0$ such that for every $x \in (0, \pi) \setminus \mathcal{R}$, a proportion at least γ of the triples (j, u, r) satisfies*

$$|b_{j,u,r}(x)| \geq c_1 A_L(x).$$

Proof. By Lemmas 4, 7 and 5,

$$\begin{aligned} \frac{1}{qL^2} \sum_{j,u,r} Y_{j,u,r}(x)^2 &\geq 4 \cdot \frac{3}{8} \cdot \frac{1}{L} \sum_{r=L}^{2L-1} \sin^2(rx/2) \\ &\geq c_0 \Delta(x)^2, \end{aligned}$$

with $c_0 = 3c_A/2 > 0$. On the other hand each term has the uniform upper bound

$$|Y_{j,u,r}(x)| \leq 4\Delta(x).$$

Indeed, when $\Delta = 1$ Lemma 4 gives $|Y| \leq 2$; when $\Delta = Ls < 1$,

$$|\sin(rx/2)| \leq r \sin(x/2) \leq 2Ls,$$

so again $|Y| \leq 4\Delta$. Here the bound $|\sin(rt)| \leq r \sin t$ for $0 \leq t \leq \pi/2$ follows by induction from the sine addition formula, since $|\sin(rt)| \leq |\sin((r-1)t)| + \sin t$.

Let $a = \sqrt{c_0/2}$. If ρ denotes the proportion of triples with $|Y_{j,u,r}| \geq a\Delta$, then

$$c_0\Delta^2 \leq 16\rho\Delta^2 + \frac{c_0}{2}(1-\rho)\Delta^2,$$

and therefore

$$\rho \geq \frac{c_0/2}{16 - c_0/2} =: \gamma > 0.$$

For these triples,

$$|b_{j,u,r}| = \frac{|Y_{j,u,r}|}{2s} \geq \frac{a}{2} \frac{\Delta}{s} = \frac{a}{2} \min\{L, 1/s\}.$$

Taking $c_1 = a/2$ proves the lemma. $\square$

Lemma 9 (Abundance of good blocks in the random boxes). *There are absolute constants $c_2, c_3 > 0$ such that for every $x \in (0, \pi) \setminus \mathcal{R}$,*

$$\Pr_{U,R}(\#\{j : |b_j(x)| \geq c_1 A_L(x)\} < c_2 q) \leq e^{-c_3 q},$$

where $b_j(x) = b_{j,U_j,R_j}(x)$.

Proof. Let X_j be the indicator of the event that the j -th random box satisfies $|b_j(x)| \geq c_1 A_L(x)$, and write $p_j = \mathbb{E}X_j$. Lemma 8 is exactly the statement

$$\sum_{j=0}^{q-1} p_j \geq \gamma q.$$

The X_j are independent, though not necessarily identically distributed. Let $S = \sum_j X_j$ and $\mu = \mathbb{E}S = \sum_j p_j$. Taking $\lambda = \log 2$ and applying the exponential Markov inequality together with independence,

$$\begin{aligned} \Pr(S \leq \mu/2) &\leq e^{\lambda\mu/2} \mathbb{E}e^{-\lambda S} \\ &= e^{\lambda\mu/2} \prod_j (1 - p_j + p_j e^{-\lambda}) \\ &\leq \exp\left[\frac{\lambda\mu}{2} - (1 - e^{-\lambda})\mu\right] \\ &\leq e^{-\mu/8} \leq e^{-\gamma q/8}. \end{aligned}$$

Taking $c_2 = \gamma/8$ and $c_3 = \gamma/8$, and noting that $c_2 q \leq \mu/2$, proves the lemma. Observe that the argument uses only independence; identical distribution is never invoked. $\square$

7 A weighted Littlewood–Offord estimate

Lemma 10 (Small-ball estimate with arbitrary centre). *Let $\eta_1, \dots, \eta_r$ be independent Bernoulli(1/2) random variables, and let $a_1, \dots, a_r$ be real weights with $|a_i| \geq a > 0$. Then for every $t \in \mathbb{R}$ and $w \geq 0$,*

$$\Pr\left(\left|\sum_{i=1}^r \eta_i a_i - t\right| \leq w\right) \leq \frac{C(1 + w/a)}{\sqrt{r}}.$$

Proof. If $a_i < 0$, replace η_i by $1 - \eta_i$ and absorb the resulting constant into the centre t . We may therefore assume $a_i \geq a$ for all i .

First consider any half-open interval J with $|J| = a$. The family of subsets whose sums lie in J is an antichain: if $S \subsetneq T$, then

$$\sum_{i \in T} a_i - \sum_{i \in S} a_i = \sum_{i \in T \setminus S} a_i \geq a,$$

so two such sums cannot both lie in one half-open interval of length a . By Sperner’s theorem the probability is at most

$$2^{-r} \binom{r}{\lfloor r/2 \rfloor} \leq C/\sqrt{r}.$$

Now cover the closed interval $[t - w, t + w]$ by $O(1 + w/a)$ half-open intervals of length a , plus at most one endpoint singleton. The subsets whose sums land on the singleton also form an antichain. The union bound over these pieces yields the lemma. $\square$

For completeness we record the self-contained proof of Sperner’s theorem used above. A maximal chain is produced by a permutation, adding one element at a time in the order prescribed by the permutation. There are $r!$ such chains, and a set of size k lies in exactly $k!(r - k)!$ of them. If $\mathcal{F}$ is an antichain, each maximal chain contains at most one member of $\mathcal{F}$. Double-counting the set–chain incidences gives

$$\sum_{S \in \mathcal{F}} \frac{1}{\binom{r}{|S|}} \leq 1.$$

Since $\binom{r}{|S|} \leq \binom{r}{\lfloor r/2 \rfloor}$, we obtain $|\mathcal{F}| \leq \binom{r}{\lfloor r/2 \rfloor}$. The central binomial estimate $\binom{r}{\lfloor r/2 \rfloor} \leq C2^r/\sqrt{r}$ is immediate from Stirling’s formula.

Corollary 11 (Small-ball probability for a random block sum). *Fix $x \notin \mathcal{R}$. For the random polynomial defined in Section 4,*

$$\Pr(x \in E(g)) \leq \begin{cases} C \min\{1, (xL\sqrt{q})^{-1} + q^{-1/2}\} + e^{-cq}, & 0 < x \leq 1/L, \\ Cq^{-1/2} + e^{-cq}, & 1/L < x < \pi. \end{cases}$$

Proof. First fix the box geometry. On the good event of Lemma 9, at least $c_2 q$ of the weights satisfy

$$|b_j(x)| \geq c_1 \min\{L, 1/s\}.$$

The condition $x \in E(g)$ is equivalent to

$$\left|\sum_j \eta_j b_j(x) - \frac{1}{2}\right| \leq \frac{1}{2s}.$$

Apply Lemma 10 to these good weights, conditioning on the remaining η_j . The ratio of the window to the good-weight scale is

$$\frac{1/(2s)}{c_1 \min\{L, 1/s\}} \ll \begin{cases} 1/(Lx), & 0 < x \leq 1/L, \\ 1, & 1/L < x < \pi, \end{cases}$$

where we used $x/\pi \leq \sin(x/2) \leq x/2$. Finally add the geometric bad event e^{-cq} and truncate the probability by 1. $\square$

8 Expected measure of the small-band set

Proposition 12. *The random construction of Section 4 satisfies*

$$\mathbb{E} |E(g)| \ll \frac{1}{q} + \frac{\log q}{L\sqrt{q}} + \frac{1}{\sqrt{q}}.$$

Proof. Since the indicator functions are nonnegative, Tonelli's theorem gives

$$\mathbb{E} |E(g)| = \int_0^\pi \Pr(x \in E(g)) dx.$$

The point is that the probability is taken at each fixed x before integrating, so there is no need to build a single high-probability event valid simultaneously for uncountably many x .

On the resonance layer $\mathcal{R}$ we use the trivial probability bound 1; by Lemma 6 its contribution is $O(1/q)$. On the complement set

$$x_0 = (L\sqrt{q})^{-1}.$$

On $(0, x_0)$ we use the trivial bound 1, contributing x_0 . On $(x_0, 1/L)$ we integrate the $1/x$ term of Corollary 11, obtaining

$$\frac{C}{L\sqrt{q}} \int_{x_0}^{1/L} \frac{dx}{x} = \frac{C \log \sqrt{q}}{L\sqrt{q}}.$$

The remaining contributions are $O(q^{-1/2})$, and the term e^{-cq} is absorbed into the same expression. Combining these estimates proves the proposition. $\square$

Corollary 13. *If $q \asymp m/L$ and $L \asymp \log m$, then there exists a deterministic 0/1 cosine polynomial g satisfying*

$$|E(g)| \ll \sqrt{\frac{\log m}{m}}.$$

Proof. Substituting $q \asymp m/L$ into Proposition 12 gives

$$\mathbb{E} |E(g)| \ll \frac{L}{m} + \frac{\log(m/L)}{\sqrt{mL}} + \sqrt{\frac{L}{m}}.$$

Taking $L = \lfloor \log m \rfloor$, all three terms are $\leq C\sqrt{\log m/m}$. By the averaging principle, at least one complete realization satisfies the stated bound. $\square$

9 Deterministic zero counting

Lemma 14 (Zeros of a trigonometric polynomial). *A nonzero real trigonometric polynomial*

$$P(x) = \sum_{|k| \leq d} c_k e^{ikx}$$

has at most $2d$ distinct zeros in one period.

Proof. Set $z = e^{ix}$ and multiply $P(x)$ by z^d . The resulting expression is an ordinary polynomial of degree at most $2d$. If it had more than $2d$ distinct roots on the unit circle, it would vanish identically, forcing $P \equiv 0$, a contradiction. $\square$

Polynomials with half-integer frequencies have $O(d)$ zeros as well: writing $y = x/2$ reduces them to integer-frequency trigonometric polynomials in y .

Theorem 15 (Deterministic counting). *Let g be a $0/1$ cosine polynomial of degree at most m , and let $n > m$. Then*

$$Z(D_n - g) \leq C(n|E(g)| + m + 1).$$

Proof. Let h, M be as in Section 3, and define on the half-period

$$U = \{x \in (0, \pi) : |h(x)| < 1\}.$$

Step 1: the number of components of U . The function h is a sine polynomial with half-integer frequencies and highest frequency at most $m + 1/2$. Hence $h^2 - 1$ is an integer-frequency trigonometric polynomial of degree at most $2m + 1$. It is not identically zero because $h(0) = 0$. By Lemma 14 the equation $|h| = 1$ has only $O(m + 1)$ solutions, so U has only $O(m + 1)$ connected components.

Step 2: total variation of the arcsine phase. On each component of U define

$$\alpha(x) = \arcsin h(x) \in (-\pi/2, \pi/2).$$

The derivative h' is a half-integer-frequency trigonometric polynomial of degree $O(m)$; unless it vanishes identically, it has only $O(m + 1)$ zeros. Subdivide U by the boundary points and the zeros of h' ; this yields $O(m + 1)$ intervals in total. On each such interval h is monotone, hence so is α , and the variation of α on a single interval is at most π . Therefore

$$\text{Var}_U(\alpha) = O(m + 1).$$

If $h' \equiv 0$, then $h \equiv 0$ (as $h(0) = 0$), and the conclusion is immediate.

Step 3: writing zeros as two families of lattice hits. Inside U , the equation $\sin(Mx) = h(x)$ is equivalent to

$$\psi_-(x) := Mx - \alpha(x) \in 2\pi\mathbb{Z},$$

or

$$\psi_+(x) := Mx + \alpha(x) \in (2\mathbb{Z} + 1)\pi.$$

The interior critical points of both phase functions satisfy

$$P(x) := M^2(1 - h(x)^2) - h'(x)^2 = 0.$$

The function P is an integer-frequency trigonometric polynomial of degree at most $2m + 1$. It is not identically zero: indeed

$$h'(0) = g(0) - \frac{1}{2}, \quad |h'(0)| \leq m + \frac{1}{2} < M,$$

where the last inequality uses $n > m$; hence

$$P(0) = M^2 - h'(0)^2 > 0.$$

By Lemma 14, the equation $P = 0$ has only $O(m + 1)$ solutions.

Step 4: counting lattice points on monotone pieces. Subdivide U by the boundary points and the zeros of P . On each resulting interval, ψ_- and ψ_+ have no critical points and are therefore strictly monotone. A monotone function hits lattice points of spacing 2π at most

$$1 + \frac{\text{variation on the interval}}{2\pi}$$

times. Summing over all intervals and using

$$\text{Var}_U(\psi_{\pm}) \leq M|U| + \text{Var}_U(\alpha),$$

the number of zeros of $D_n - g$ lying in U on $(0, \pi)$ is at most

$$C(n|U| + m + 1).$$

The boundary $|h| = 1$ contains only $O(m + 1)$ points, and no zero can occur where $|h| > 1$. Hence the same bound controls the whole half-period. Finally, using

$$D_n(2\pi - x) - g(2\pi - x) = D_n(x) - g(x),$$

the half-period result is multiplied by 2, with only endpoint constants added. Since $E(g)$ and U differ by finitely many boundary points, $|U| = |E(g)|$. This proves the theorem. $\square$

Remark 16 (On multiplicities). The lattice-hit argument above counts *distinct* zeros, which matches the definition of $Z(N)$. If one insisted on counting zeros with multiplicity, one could not simply quote “a monotone function hits lattice points”; one would have to add the extra multiplicities of interior roots to the multiplicities of the zeros of the critical polynomial P , and to perform a separate local expansion on the boundary $|h| = 1$. The audit confirms that this strengthening can be carried out, but the main theorem does not require it.

10 Exact term count and parameter balancing

Fix a realization g as in Corollary 13, and let K be the number of its coefficients equal to 1. Since every block has length less than $2L$,

$$0 \leq K \leq 2Lq \leq \frac{m+1}{2}.$$

Given the target number of terms N , define after the fact

$$n = N + K - 1.$$

Since the support of g is contained in $\{0, \dots, m\}$ and the final parameters satisfy $n > m$, the polynomial $D_n - g$ still has 0/1 coefficients and has exactly

$$(n + 1) - K = N$$

nonzero terms.

Now choose

$$m = \left\lceil N^{2/3}(\log N)^{1/3} \right\rceil, \quad L = \lfloor \log m \rfloor.$$

As $N \rightarrow \infty$,

$$\frac{m}{N} = N^{-1/3}(\log N)^{1/3}(1 + o(1)) \rightarrow 0. \quad (1)$$

Hence for sufficiently large N we have $n \geq N - 1 > m$, and, by the bound on K ,

$$n \leq N + \frac{m + 1}{2} \leq 2N.$$

Applying Theorem 15 and Corollary 13,

$$\begin{aligned} Z(N) &\leq Z(D_n - g) \leq C \left(n \sqrt{\frac{\log m}{m}} + m + 1 \right) \\ &\ll N^{2/3}(\log N)^{1/3}. \end{aligned}$$

This proves Theorem 1.

Remark 17 (Degenerate and boundary cases). To avoid extrapolating the asymptotic argument to regimes where it does not apply, we record the boundary conditions explicitly. For $N = 1$ one may take $A = \{0\}$, whose polynomial is identically 1 and has no zeros. For finitely many small N the random construction is not used; the finite maximum is absorbed into the absolute constant. The conditions $L < 8$ or $q < 8$ occur only for finitely small parameters, and the probabilistic argument begins at $L, q \geq 8$. At $x = 0$ one has $D_n(0) - g(0) = N > 0$, which is not a zero and does not affect the Tonelli integral. The point $x = \pi$ lies in the resonance layer, so it has measure zero and the half-period endpoint contributes only a constant. The resonant points $x = k\pi/(4L)$ are all placed inside $\mathcal{R}$, so the non-resonant phase average is never applied to them. Finally, the condition $n > m$ in Theorem 15 cannot be dropped: if $g = D_m$, then $D_n - g \equiv 0$; and by (1) one has $m = o(N)$, so all frequency-separation conditions eventually hold.

A Conditions that cannot be removed

The following statements, each of which might appear to be a harmless strengthening, are false, and were eliminated during the audit. They are recorded here so that they are not reintroduced into the proof.

1. *A per-block lower bound with an arbitrary extra phase.* Taking $\theta = \pi/2$, $x \rightarrow 0$, $Lx \rightarrow 0$, the left-hand side is $O(L^4 x^4)$ while the required lower bound is $\asymp (Lx)^2$. The argument avoids this failure by discarding the resonance layer $\mathcal{R}$.
2. *Subset sums inside a closed interval of length a form an antichain.* The single weight a with the closed interval $[0, a]$ is a counterexample. The proof uses a partition into half-open intervals instead.

3. *Writing $|\mathcal{R}| = 2 \arcsin(D/q)$ for all q .* When $D/q > 1$ the expression is not real-valued; the correct formula is Lemma 6.
4. *Dropping the condition $n > m$.* As noted in Remark 17, $n = m$ with $g = D_m$ makes the difference polynomial identically zero.

AI disclosure statement

ChatGPT 5.6 Sol by OpenAI was used to explore proof strategies for this note.

References

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